

Lab: Sequential Decisions -- Thinking Multiple Steps Ahead

Objective

Practice **multi-step decision analysis** using decision trees, expected values across time, and the concept of **expected free energy**. In Active Inference, planning means evaluating policies (sequences of actions) by computing their expected free energy over future time steps.

Prerequisites

- Completed Math Foundations: Communication module (entropy, information gain)
- Expected value calculations, basic tree diagrams

Part 1: Decision Trees

Goal: Evaluate a two-step decision by computing expected values at each branch.

You have a project due in 2 weeks. At week 1, you choose: Start now or Wait.

If you Start now: - P(finish early) = 0.7, value = 10 - P(need rush) = 0.3, value = 5

If you Wait: - At week 2, you can Work hard or Wing it - Work hard: P(decent grade) = 0.6, value = 7; P(poor grade) = 0.4, value = 3 - Wing it: P(decent grade) = 0.2, value = 7; P(poor grade) = 0.8, value = 3

1. Draw the decision tree with all branches labeled.
2. $E[\text{Start now}] = ?$
3. $E[\text{Wait, then Work hard}] = ?$
4. $E[\text{Wait, then Wing it}] = ?$
5. If you Wait, which week-2 action is better? What is $E[\text{Wait}]$ using the best week-2 choice?
6. Compare $E[\text{Start now}]$ vs. $E[\text{Wait}]$. What is the optimal policy (sequence of decisions)?

Write your response here

Part 2: Discounting Future Rewards

Goal: Understand how agents value near vs. distant outcomes.

A discount factor γ ($0 < \gamma < 1$) reduces the value of future rewards: $V = r_{\text{now}} + \gamma * r_{\text{next}} + \gamma^2 * r_{\text{later}}$.

Policy A: rewards = [2, 2, 2] (steady) Policy B: rewards = [0, 0, 8] (delayed payoff)

1. With $\gamma = 1.0$ (no discounting): $V(A) = ?$, $V(B) = ?$
2. With $\gamma = 0.5$: $V(A) = 2 + 0.5(2) + 0.25(2) = ?$, $V(B) = ?$
3. With $\gamma = 0.9$: $V(A) = ?$, $V(B) = ?$
4. At what discount factor does Policy A become better than Policy B?
5. In Active Inference, the time horizon of planning is related to how far into the future the agent evaluates expected free energy.

Write your response here

Part 3: Expected Free Energy (Simplified)

Goal: Combine expected reward and expected information gain into a single score.

Recall: Expected Free Energy (G) = Expected Loss + Expected Uncertainty (negative of: pragmatic value + epistemic value).

Two study policies for a test next week:

Policy 1 (Review known material): - Expected grade: 85/100 (pragmatic value = 85) - Expected information gain: low (epistemic value = 5)

Policy 2 (Practice new problems): - Expected grade: 75/100 (pragmatic value = 75) - Expected information gain: high (epistemic value = 25)

1. $\text{Score}(\text{Policy 1}) = \text{pragmatic} + \text{epistemic} = ?$
2. $\text{Score}(\text{Policy 2}) = \text{pragmatic} + \text{epistemic} = ?$
3. Which policy has lower expected free energy (higher combined score)?
4. Why might an Active Inference agent prefer Policy 2 early in studying but switch to Policy 1 the night before the test?

Write your response here

Part 4: Policy Trees and Evaluation

Goal: Enumerate and compare complete policies.

A robot agent at position A can move to B or C, then from there to D or E.

Policy	Step 1	Step 2	P(success)	Reward if success	Info gained
pi_1	A->B	B->D	0.8	10	2
pi_2	A->B	B->E	0.5	15	5
pi_3	A->C	C->D	0.6	12	3
pi_4	A->C	C->E	0.3	20	8

1. Compute the expected reward for each policy: $E[\text{reward}] = P(\text{success}) * \text{reward}$.
2. Compute a combined score for each: $\text{Score} = E[\text{reward}] + \text{info_gained}$.

3. Rank the policies by expected reward alone. Then rank by combined score. Does the ranking change?
4. In Active Inference, the agent assigns a probability to each policy proportional to $\exp(-G)$, where G is the expected free energy. Policies with lower G (better combined score) are more likely to be selected.

Write your response here

Summary Table

Math Concept	Symbol	Definition
Decision tree	--	Branching diagram of sequential choices and outcomes
Discount factor	gamma	Reduces the weight of future rewards ($0 < \text{gamma} < 1$)
Expected free energy	G	Combines expected loss and expected uncertainty over future steps
Policy	pi	A complete sequence of planned actions
Policy evaluation	$E[G(\text{pi})]$	Scoring a policy by its expected free energy across time steps